2.Trial Run (JOG run)

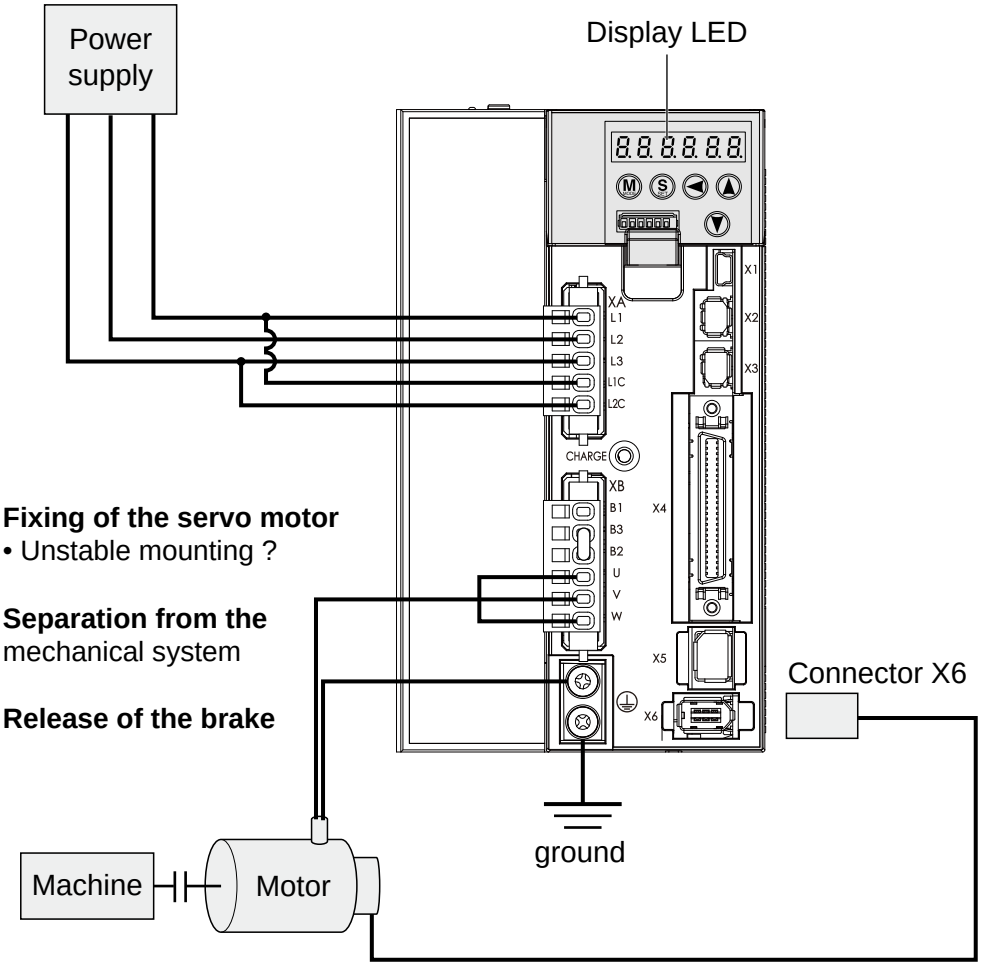
Inspection Before Trial Run

(1) Inspection on wiring

- Miswiring ? (Especially power input and motor output)
- Short or grounded ?
- Loose connection ?

(2) Confirmation of power supply and voltage

- Rated voltage ?



(3) Fixing of the servo motor

- Unstable mounting ?

(4) Separation from the mechanical system

(5) Release of the brake

(6) Turn to Servo-OFF after finishing the trial run by pressing (S SET).

Note Details of wiring, refer to P.2-8 “Overall Wiring”

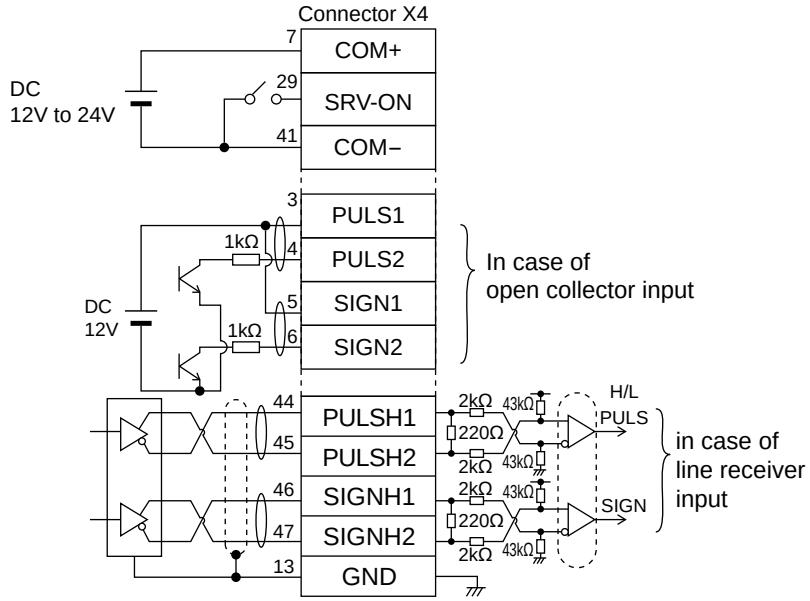
2.Trial Run (JOG run)

Trial Run by Connecting the Connector, CN X4

Trial Run (JOG run) at Position Control Mode

(1) Connect the Connector X4.
(2) Enter the power (DC12 to 24V) to control signal (COM+, COM–)
(3) Enter the power to the driver.
(4) Confirm the default values of parameters.
(5) Match to the output format of the host controller with Pr0.07 (Command pulse input mode setup).
(6) Write to EEPROM and turn off/on the power (of the driver).
(7) Connect the Servo-ON input (SRV-ON, Connector X4, Pin-29) and COM– (Connector X5, Pin-41) to bring the driver to Servo-ON status and energize the motor.
(8) Enter low frequency from the host controller to run the motor at low speed.
(9) Check the motor rotational speed at monitor mode whether,
rotational speed is as per the setup or not, and
the motor stops by stopping the command (pulse) or not.
(10) If the motor does not run correctly, refer to P.2-64, "Display of Factor for No-Motor Running" of Preparation.

• Wiring Diagram



• Parameter

Pr No.	Title	Setup value
0.01	Control mode setup	0
5.04	Over-travel inhibit input setup	1
0.05	Selection of command pulse input	0/1
0.07	Command pulse input mode setup	1
5.18	Invalidation of command pulse inhibit input	1
5.17	Counter clear input mode	2

• Input signal status

No.	Title of signal	Monitor display
0	Servo-ON	+A

Trial Run (JOG run) at Velocity Control Mode

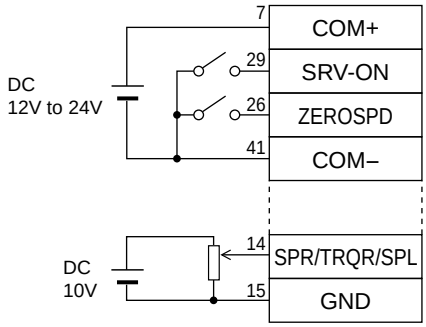
1) Connect the Connector X4.
2) Enter the power (DC12 to 24V) to control signal (COM+, COM–)
3) Enter the power to the driver.
4) Confirm the default values of parameters.
5) Connect the Servo-ON input (SRV-ON, Connector X4, Pin-29) and COM– (Connector X4, Pin-14) to turn to Servo-ON and energize the motor.
6) Close the speed zero clamp input (ZEROSPD) and apply DC voltage between velocity command input , SPR (Connector X4, Pin-14) and GND (Connector X4, Pin-15), and gradually increase from 0V to confirm the motor runs.
7) Confirm the motor rotational speed in monitor mode.
   - Whether the rotational speed is per the setup or not.
   - Whether the motor stops with zero command or not.
8) If the motor does rotate at a micro speed with command voltage of 0.
9) When you want to change the rotational speed and direction, set up the following parameters again.

   Pr3.00: Speed setup, Internal/External switching
   Pr3.01: Speed command rotational direction selection
   Pr3.03: Reversal of speed command input

   Refer to P.4-24, "Parameter Setup" (Parameters for Velocity/Torque Control)

10) If the motor does not run correctly, refer to P.2-64, "Display of Factor for No-Motor Running" of Preparation.

• Wiring Diagram



Run with ZEROSPD switch close, and Stop with open

In case of bi-directional operation (Positive/Negative), provide a bipolar power supply, or use with Pr03.15 = 3.

In case of one-directional operation

• Parameter

Pr No.	Title	Setup value
0.01	Control mode setup	1
5.04	Over-travel inhibit input setup	1
3.15	Speed zero-clamp function selection	1
3.00	Speed setup, Internal/External switching	Set up as required
3.01	Speed command rotational direction selection	
3.02	Input gain of speed command	
3.03	Reversal of speed command input	
4.22	Analog input 1 (AI1) offset setup	
4.23	Analog input 1 (AI1) filter	

• Input signal status

No.	Title of signal	Monitor display
0	Servo-ON	+A
5	Speed zero clamp	—

Trial Run (JOG run) at Torque Control Mode

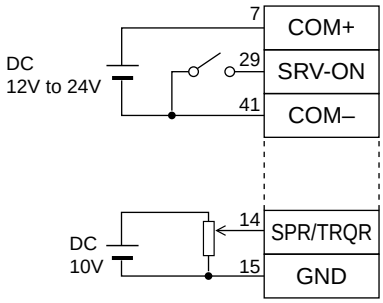
1) Connect the Connector X4.
2) Enter the power (DC12-24V) to control signal (COM+, COM–)
3) Enter the power to the driver.
4) Confirm the default values of parameters.
5) Set a lower value to Pr3.07 (4th speed of speed setup).
6) Energize the motor by connecting the Servo-ON input (SRV-ON, Connector X4, Pin-29) and COM– (Pin-41 of Connector X4) to turn to Servo-ON status.
7) Confirm that the motor runs as per the setup of Pr3.07 by applying DC voltage (positive/negative) between the torque command input (Pin-14 of Connector X4) and GND (Pin-41 of Connector X4).
8) If you want to change the torque magnitude, direction and velocity limit value against the command voltage, set up the following parameters.

Pr3.19: Input gain of torque command
Pr3.20: Input reversal of torque command
Pr3.21: Speed limit value 1

Refer to P.2-28, 29, "Parameter Setup" (Parameters for Velocity/Torque Control)

9) If the motor does not run correctly, refer to P.2-64, "Display of factor for No-motor running" of Preparation.

• Wiring Diagram



For bi-directional running (Positive/Negative), provide a bipolar power supply.

In case of one way running

• Parameter

Pr No.	Title	Setup value
0.01	Control mode setup	2
5.04	Over-travel inhibit input setup	1
3.15	Speed zero-clamp function selection	0
3.17	Selection of torque command	0
3.19	Input gain of torque command	Set up as required
3.20	Input reversal of torque command	
3.21	Speed limit value 1	lower value

• Input signal status

No.	Title of signal	Monitor display
0	Servo-ON	+A
5	Speed zero clamp	—

2.Trial Run (JOG run)

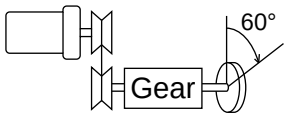
Setup of Motor Rotational Speed and Input Pulse Frequency

Input pulse frequency (pps)	Motor rotational speed (r/min)	Pr0.08	
		17-bit	20-bit
2M	3000	$\frac{2^{17}}{40000}$	$\frac{2^{20}}{40000}$
500K	3000	$\frac{2^{17}}{10000}$	$\frac{2^{20}}{10000}$
250K	3000	$\frac{2^{17}}{5000}$	$\frac{2^{20}}{5000}$
100K	3000	$\frac{2^{17}}{2000}$	$\frac{2^{20}}{2000}$
500K	1500	$\frac{2^{17}}{20000}$	$\frac{2^{20}}{20000}$

Note When setting Pr0.08, and encoder resolution is automatically set up as numerators.

Caution
- Max. input pulse frequency varies depending on input terminals.
- You can set up any values to numerator and denominator, however, setup of an extreme division ratio or multiplication ratio may result in dangerous action. Recommended ratio is 1/1000 to 1000.

Relation between the motor rotational speed and input pulse counts



Pulley ratio : $\frac{18}{60}$

Gear ratio : $\frac{12}{73}$

Total reduction ratio : $\frac{18}{365}$

When setting the command division and multiplication ratio as numerator/denominator, express it as Pr0.09/Pr0.10 with Pr0.08 = 0.

e.g.) When you want to rotate the motor by 60° with the load of total reduction ratio of 18/365.

	Encoder	
	17-bit	20-bit
$\frac{\text{Pr0.09}}{\text{Pr0.10}}$	$\frac{5840}{108}$	$\frac{5840}{67500}$
Command pulse	To rotate the output shaft by 60°, enter the command of 8192 (2^{13}) pulses from the host controller.	To rotate the output shaft by 60°, enter the command of 10000 pulses from the host controller.
How to determine parameter	$\frac{365}{18} \times \frac{1\times2^{17}}{2^{13}} \times \frac{60°}{360°} = \frac{5840}{108}$	$\frac{365}{18} \times \frac{1\times2^{20}}{10000} \times \frac{60°}{360°} = \frac{5840}{67500}$

2^n	Decimal figures
2^0	1
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128
2^8	256
2^9	512
2^{10}	1024
2^{11}	2048
2^{12}	4096
2^{13}	8192
2^{14}	16384
2^{15}	32768
2^{16}	65536
2^{17}	131072
2^{18}	262144
2^{19}	524288
2^{20}	1048576

*Refer to P.2-50 "Division Ratio for Parameters" of Supplement.